Evaluation of NC-FinFET Based Subsystem-Level Logic Circuits

Wei-Xiang You, Pin Su, *Member, IEEE*, and Chenming Hu, *Life Fellow, IEEE*

***Abstract*— This paper examines metal–ferroelectric–insulator–semiconductor negative-capacitance FinFET (NC-FinFET) based VLSI subsystem-level logic circuits. For the first time, with the aid of a short-channel NC-FinFET compact model, we confirm the functionality and evaluate the standby-power/switching-energy/delay performance of large logic circuits (e.g., dynamic 4-bit Manchester carry-chain adder and the formal hierarchical 32-bit carry-look-ahead adder) employing 14-nm ultra-low-power NC-FinFETs. Our study indicates that the inverse V_{ds}-dependence of threshold voltage (V_T), also known as the negative drain-induced barrier lowering, of negative-capacitance field-effect transistor is not only acceptable but also beneficial for the speed performance of both the static and pass-transistor logic (PTL) circuits, especially for the PTL at low V_{DD}.**

***Index Terms*— Dynamic adder, FinFET, Landau–Khalatnikov (L–K) equation, logic circuits, metal–ferroelectric–insulator–semiconductor (MFIS)-type negative-capacitance field-effect transistor (NCFET), NCFET, pass-transistor logic (PTL).**

I. Introduction

FOR meeting the high-speed and low-power requirement of VLSI, negative-capacitance field-effect transistor (NCFET) is one of the most promising device candidates. It can achieve a subthreshold swing (SS) smaller than 60 mV/decade at room temperature [1]–[6]. Using a production 14-nm FinFET technology platform, NC-FinFET ring oscillators operating at high frequencies have recently been reported [3], thus providing encouraging evidence of the intrinsic speed [7] (~7.2 ps per stage delay [8]) and endurance of the negative-capacitance (NC) effect [3]. In addition to the improved SS and ON-state current (I_{ON}), the distinctive inverse drain-bias (V_{ds})-dependence of threshold voltage (V_T) characteristic [also known as the negative drain-induced barrier lowering (DIBL)] has been observed in the NC-FinFET hardware [6], [9]. In [10], the delay and switching energy of an inverter with metal–ferroelectric–metal–insulator–semiconductor-type NC-FinFETs have been evaluated. However, the functionality and performance gain of large VLSI circuits employing NC-FinFETs, especially NC-FinFETs with strong inverse V_{ds}-dependence of V_T have yet to be investigated.

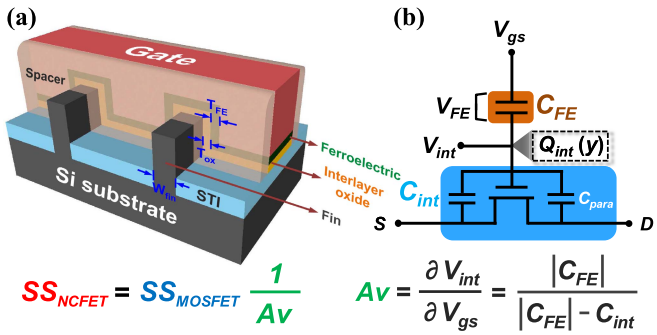


Fig. 1. (a) Schematic of an MFIS-type NC-FinFET with T_{ox}, T_{FE}, and W_{fin} the interlayer oxide thickness, ferroelectric thickness, and fin width, respectively. (b) Equivalent circuit representation of the MFIS-type NCFET with C_{FE} the ferroelectric capacitance, C_{int} the internal capacitance, C_{para} the parasitic capacitance, V_{FE} the voltage drop across the ferroelectric, and V_{int} the internal voltage.

In this paper, with the aid of a developed short-channel NC-FinFET SPICE model compatible with Berkeley Short-channel IGFET Model-Common Multi-Gate (BSIM-CMG) [11], we examine the functionality and evaluate the standby-power/switching-energy/delay performance of VLSI subsystem-level logic circuits (such as dynamic adders) using metal–ferroelectric–insulator–semiconductor (MFIS) NC-FinFETs. The impacts of the unique inverse V_{ds}-dependence of V_T and negative output conductance on both the static, dynamic, and pass-transistor logic (PTL) circuits are highlighted.

This paper is organized as follows. In Section II, the methodology to evaluate the MFIS-type NC-FinFET based logic circuits is described. In Section III, to provide an insight into the logic circuits with NC-FinFETs, we evaluate the standby-power/switching-energy/delay performance of a five-stage inverter chain and extend our analysis to the dynamic 4-bit Manchester carry-chain (MCC) adder, 32-bit hierarchical carry-look-ahead (CLA) adder, and other types of logic circuits. In Section IV, the impact of inverse V_{ds}-dependence

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W.-X. You and P. Su are with the Department of Electronics Engineering, National Chiao Tung University, Hsinchu 30010, Taiwan, and also with the Institute of Electronics, National Chiao Tung University, Hsinchu 30010, Taiwan (e-mail: pinsu@faculty.nctu.edu.tw).

C. Hu is with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA 94720 USA, and also with the International College of Semiconductor Technology, National Chiao Tung University, Hsinchu 30010, Taiwan.

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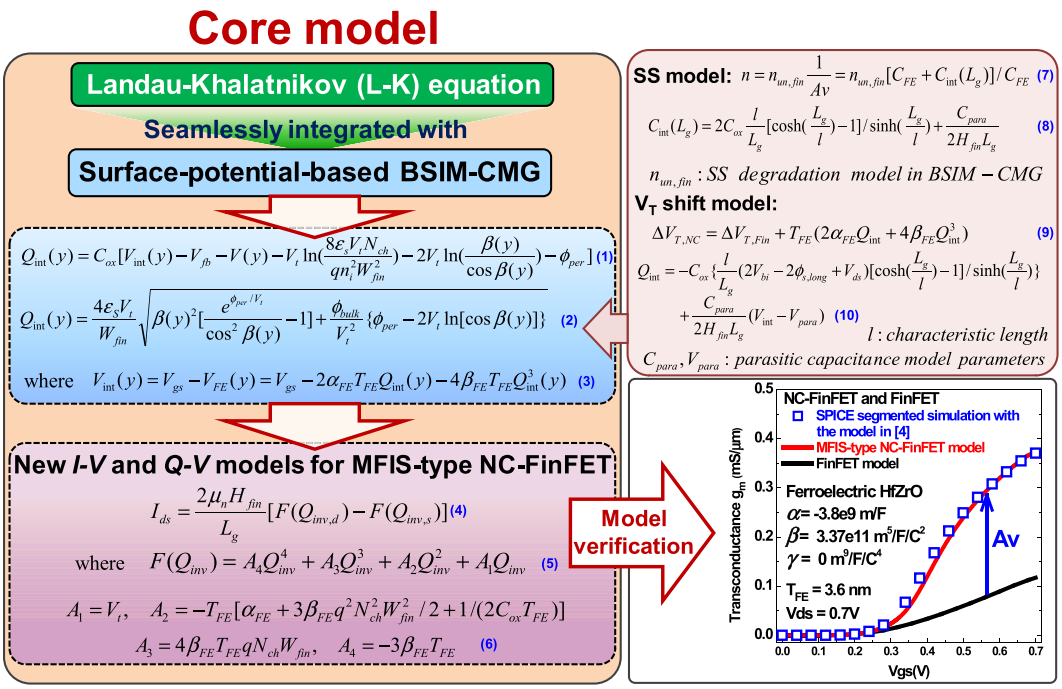


Fig. 2. Framework of our BSIM-CMG compatible NC-FinFET model. The NC effect is seamlessly integrated into the core model without the need of subcircuits. The short-channel SS and V_T shift models considering the NC effect are incorporated into our framework. Our new model can capture the MFIS-specific distributed charge effect (right).

of V_T characteristic in NC-FinFETs on several static, dynamic, and PTL circuits is investigated. Finally, the conclusion is drawn in Section V.

II. Methodology

While early NC-FinFET models assumed an internal gate for ease of modeling [10], [12], [13], the MFIS-type NC-FinFET, without an internal metal (see Fig. 1), is more practical. To perform circuit simulation and evaluation for MFIS-type NC-FinFET based VLSI subsystem-level logic circuits, a short-channel MFIS-type NC-FinFET model compatible with BSIM-CMG is developed. The model framework is illustrated in Fig. 2. In the core model, the 1-D steady-state Landau–Khalatnikov (L–K) equation has been seamlessly integrated into a unified surface potential equation (which can also be used for FinFETs) without using subcircuits. Note that different from the MFMIS-type NC-FinFET with a uniform polarization charge distribution seen by the ferroelectric, the internal polarization charge $Q_{\text{int}}(y)$ [see Fig. 1(b)] in the MFIS-type gate stack varies with the channel position y [4], [14], [15]. After the surface-potential iteration, $Q_{\text{int}}(y)$ at the interlayer oxide and ferroelectric interface can be obtained. With the surface-potential-based scheme, the MFIS-specific distributed charge effect can be naturally included in the drain current model without the need of local charge integration [4].

To capture the distinct short-channel effects (SCEs) of NCFETs including the SS improvement [16]–[18], V_T roll-up [18], and inverse V_{ds}-dependence of V_T [17]–[19] with decreasing gate length (L_g), the SS and V_T shift have been physically modeled based on an accurate short-channel Q_{int} model [see (10) in Fig. 2] and incorporated into the

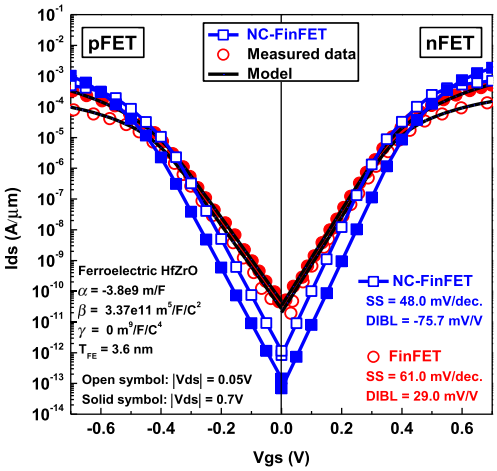


Fig. 3. I_{ds}–V_{gs} characteristics of the NC-FinFET and the 14-nm node ULP FinFET [21]. Note that $V_{T,sat} > V_{T,lin}$ for the NC-FinFET.

NC-FinFET core model. This scalable short-channel Q_{int} model [20] is derived based on an analytical short-channel surface potential equation of FinFETs [11], [18]. Then, the inverse V_{ds}-dependence of V_T characteristic can be well captured by our V_T shift model due to the inclusion of V_{ds}-dependence in the Q_{int} model [see (10) in Fig. 2]. Note that the SCEs model with the gate length dependence is capable of including both the intrinsic drain-coupling and the device parasitic capacitance.

Using the NC-FinFET model calibrated with the 14-nm node ultra-low-power (ULP) FinFET technology [21], it can be seen from Fig. 3 that the hysteresis-free I–V for NC-FinFETs with ferroelectric HfZrO and ~1.27× internal voltage gain

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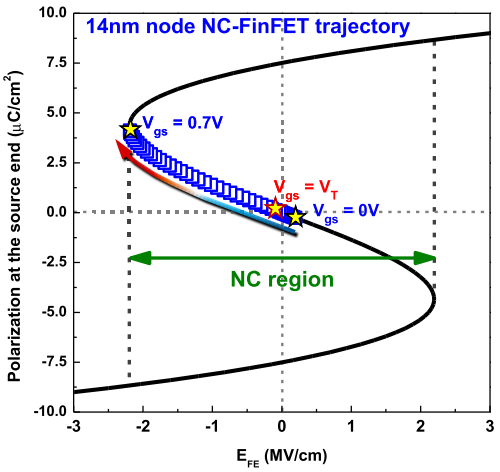


Fig. 4. Trajectory in the ferroelectric *P–E* curve demonstrating the NC-FinFET operation in the NC region.

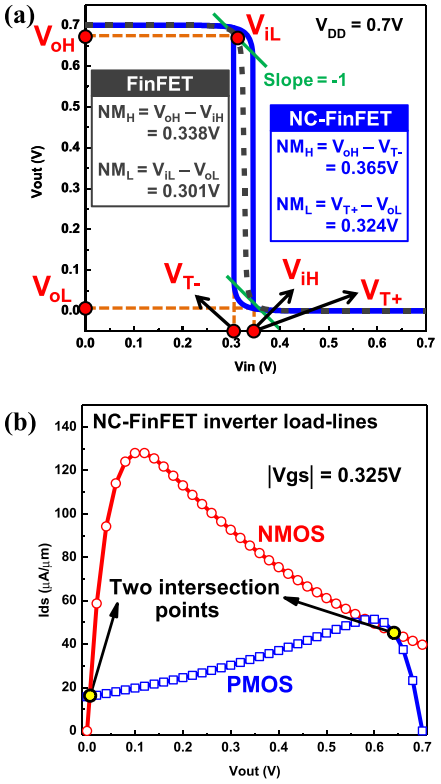


Fig. 5. (a) VTC and NMs of FinFETs and NC-FinFETs with $V_{iL/iH}$, $V_{oL/oH}$, and $V_{T-/T+}$ the low/high input logic level, low/high output logic level, and low/high threshold, respectively. The NMs of the NC-FinFET are larger than the FinFET counterparts. (b) Unique NC-FinFET VTC [Fig. 5(a)] results from the negative differential output conductance.

(the Av ranges from ∼1.1× to ∼1.3× in [3], [9], and [22]) can be achieved. The ferroelectric parameters of HfZrO used in this paper are: $\alpha = -3.8 \times 10^9$ m/F, $\beta = 3.37 \times 10^{11}$ m^5/F/C^2, and $\gamma = 0$ m^9/F/C^4. These values of α, β, and γ were obtained based on the remnant polarization (P_r) and coercive field (E_c) data of HfZrO in [23]. The model parameters for the baseline FinFET are extracted and the parasitic capacitance is also estimated (∼0.14 fF/μm) via the 3-D TCAD simulation [24]

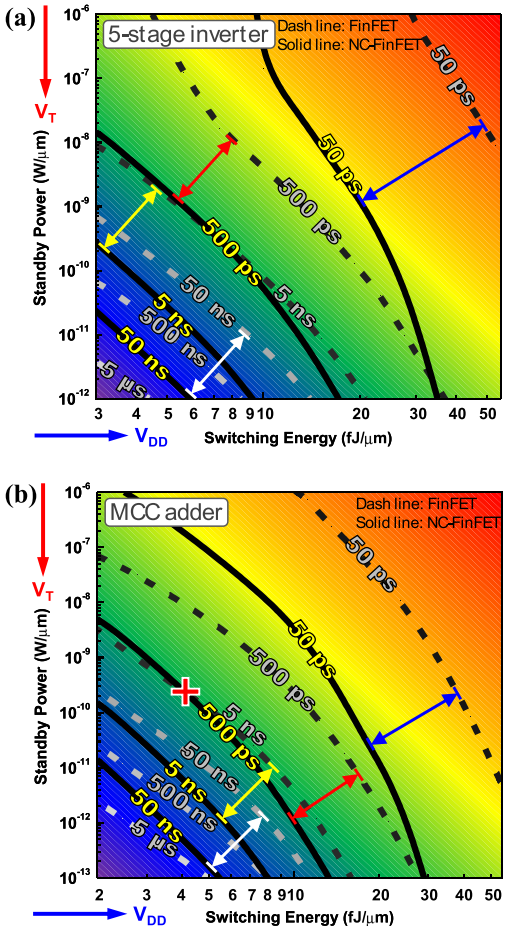


Fig. 6. Standby-power/switching-energy/delay contour plots of (a) five-stage inverter chain and (b) 4-bit dynamic MCC adder using FinFETs (dashed lines) and NC-FinFETs (solid lines). The standby-power and switching energy are varied by varying V_T and V_{DD}, respectively. For a given standby-power and switching energy [e.g., the cross in Fig. 6(b)], the speed of the MCC adder can be 10× faster with NC-FinFETs.

for the calibrated FinFET with SiO_2 spacer. With this bias-dependent Q_{int} resulted from the intrinsic drain coupling and parasitic capacitance, the ferroelectric can be polarized to enable the NC effect in the subthreshold region [15], [18] as shown in Fig. 4.

Due to the impact of NC, the improvement in SS and I_{ON} with inverse V_{ds}-dependence of V_T characteristic (i.e., $V_{T,sat} > V_{T,lin}$) can be observed for the NC-FinFET. In our circuit evaluation, the backend loading capacitance (C_{BL}) is considered based on the ITRS 14-nm technology node [25].

III. Circuits Evaluation

Even with the hysteresis-free I–V shown in Fig. 3, the NC-FinFET exhibits a voltage transfer curve (VTC) with two different logic thresholds for increasing/decreasing input signal which is different from the FinFET counterparts as shown in Fig. 5(a). This unique VTC can be explained by the inverter load lines shown in Fig. 5(b). The negative differential output conductance due to the inverse V_{ds}-dependence of V_T of the NC-FinFET leads to two intersection points for a given gate bias, and the stable solution of output voltage depends on the

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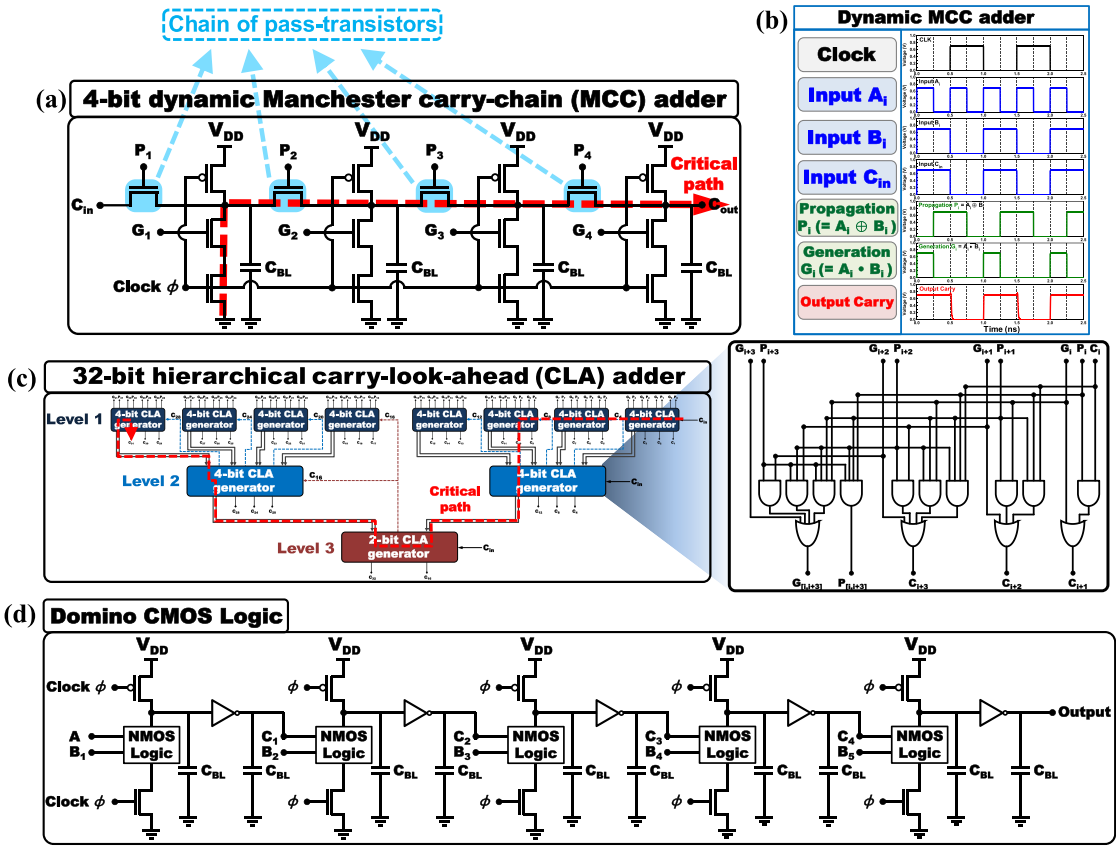


Fig. 7. Circuit schematics of (a) 4-bit dynamic MCC adder, with the waveforms showing (b) functionality of the MCC adder. (c) 32-bit hierarchical CLA adder with the gate level implementation diagram of 4-bit look-ahead carry network. (d) Domino CMOS logic. The red dashed lines in (a) and (c) represent the critical path of the adders.

direction of input voltage sweep. With this Schmitt-trigger-like VTC [26], the NC-FinFET inverter possesses larger noise margins (NMs) [no matter at the logic high-voltage (NM_H) or low-voltage (NM_L) level] than the FinFET counterparts.

Fig. 6 shows that, with V_T and supply voltage (V_{DD}) as the implicit device design variables, we introduce novel three-variable standby-power/switching-energy/delay contour plots for five-stage inverter chain and 4-bit dynamic MCC adder. Fig. 6(a) shows the comparison of a five-stage inverter chain based on FinFETs (labeled as dash lines) with one based on NC-FinFETs (labeled as solid lines). It can be seen that, for a target speed (five-stage delay) performance, the NC-FinFET inverter can operate with lower standby-power and lower switching-energy compared with the FinFET.

In addition to static logic circuits, the high-speed dynamic adder is the most extensively used component in VLSI designer library. Using the MCC adder (see Fig. 7(a), [27]–[29]) at high clock frequencies, the functionality and feasibility of performing arithmetic operation are demonstrated in Fig. 7(b). P_i/G_i is the propagation/generation signal produced by the full adders with input signals A_i, B_i, and C_{in}. Fig. 6(b) further shows the comparison of the standby-power/switching-energy/delay (critical path carry propagation delay) characteristics of the MCC adder based on FinFETs and NC-FinFETs. It can be seen that, for a given standby-power and delay, the switching-energy of the MCC adder with NC-FinFETs can be reduced by ~60% compared with the FinFET counterparts. For a given standby-power and switching-energy [see the red cross in Fig. 6(b)], the speed of the MCC adder can be 10× faster with NC-FinFETs.

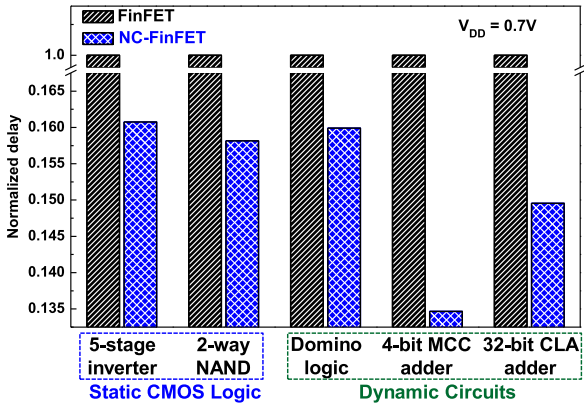


Fig. 8. Comparison of delay improvement in various types of static and dynamic logic circuits. The PG elements in the critical path of dynamic MCC [Fig. 7(a)] and CLA adders [Fig. 7(c)] are critical for the delay improvement.

Fig. 8 shows the comparison of the delay improvement in various types of static and dynamic logic circuits using NC-FinFETs under equal OFF-state current (I_{OFF}). The delay

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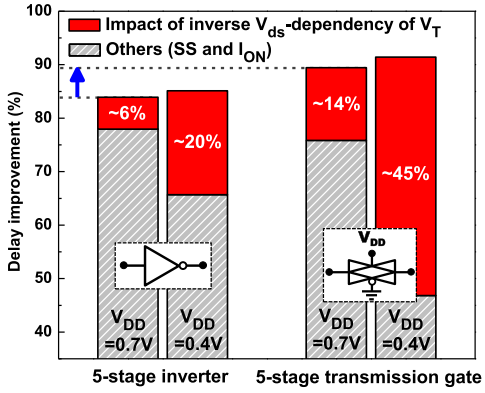


Fig. 9. Comparison of delay improvement in five-stage inverter and transmission-gate using NC-FinFETs. The contribution from the inverse V_{ds}-dependence of V_T is larger in PTL, especially at lower V_{DD} due to the subthreshold operation of NC-FinFETs.

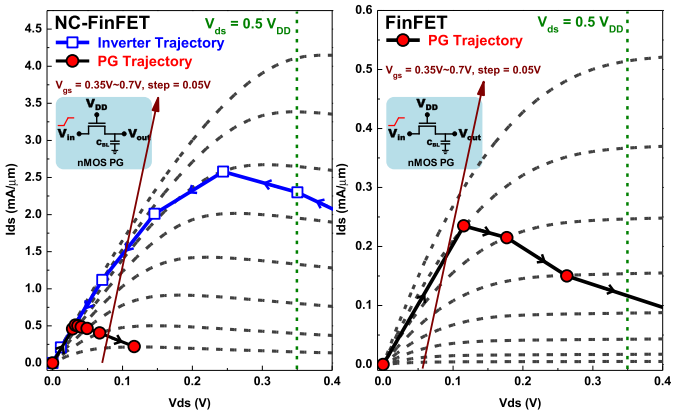


Fig. 10. Comparison of I–V trajectories of (a) NC-FinFET based PG and inverter and (b) FinFET based PG during switching showing the importance of linear region for the NC-FinFET PG.

of the static and dynamic logic circuits can be substantially improved by using NC-FinFETs. Apart from the MCC adder, high-speed adders based on the CLA principle designed to overcome the latency introduced by the rippling effect of the carry bits still remains dominant [28]–[30]. Therefore, we also evaluate formal 32-bit hierarchical CLA adder which is the representative of large dynamic circuits with the arithmetic structure shown in Fig. 7(c). The circuit implementation of the 32-bit hierarchical CLA adder is based on the modified 4-bit group pass gate (PG) and carry blocks proposed in [30]. It can be seen from Fig. 8 that more delay improvement can be observed for the dynamic 4-bit MCC and 32-bit hierarchical CLA adders compared with the other types of logic circuits (e.g., inverter, NAND, and domino logic [see Fig. 7(d)]). Note that, different from the other types of logic circuits, there exist PG elements in the critical path of the dynamic MCC and CLA adders.

IV. Impact of Inverse V_{DS}- Dependence of V_T

To clarify this delay improvement difference, we compare the FinFETs and NC-FinFETs based five-stage inverter chain and transmission-gate which are one of the typical PTL in Fig. 9. Fig. 9 shows that, in addition to the better known beneficial effects of the improved SS and I_{ON} on speed, the inverse V_{ds}-dependence of V_T of NC-FinFETs contributes significantly to the delay improvement, especially for the transmission-gate at low V_{DD}(~45% at V_{DD} = 0.4 V). This can be explained by the I–V trajectories of the NC-FinFET based PG and inverter shown in Fig. 10(a). It can be seen that quite different from the inverter, the PG trajectory lies mainly in the linear regime during switching. The degree of DIBL directly determines the discrepancy between $V_{T,lin}$ and $V_{T,sat}$ and changes the effective drive current of PG. Therefore, the performance of PTL is more sensitive to DIBL since it is determined by the low V_{ds} characteristics [31]. Note that, at a lower V_{DD}, the inverse V_{ds}-dependence of V_T plays more important role in the circuit delay improvement (see Fig. 9) due to the subthreshold operation of NC-FinFETs.

For the PTL, the output node of the PG (V_{out}) (see Fig. 10 inset) always follows the input node (V_{in}) during switching. As V_{out} is being charged, the smaller voltage drop between V_{in} and V_{out} (i.e., the drain bias of the PG) is beneficial for the NC-FinFET with a smaller $V_{T,lin}$ (see Fig. 3) so the V_{out} can be charged even faster and further reduces the voltage drop between V_{in} and V_{out}. That is why the PG with NC-FinFET operates at a lower V_{ds} regime compared with the FinFET counterpart. Due to this positive feedback in PTL with NC-FinFETs, the dynamic MCC and CLA adders with PG elements in their critical path can exhibit larger delay improvement compared with the other types of logic circuits as shown in Fig. 8.

V. Conclusion

Enabled by a physically accurate short-channel BSIM-CMG compatible MFIS-type NC-FinFET model, the functionality and the performance of VLSI subsystem-level logic circuits employing NC-FinFETs are comprehensively evaluated and analyzed for the first time. Based on our explored three-variable standby-power/switching-energy/delay contour plots, the switching energy of the high-speed adders with NC-FinFETs can be reduced by ~60% compared with the FinFET counterpart for a given standby-power and delay. Our study indicates that, in addition to the steep slope and I_{ON}, the unique inverse V_{ds}-dependence of V_T characteristic of NC-FinFETs is beneficial to the performance of logic circuits, especially for the PTL style.

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